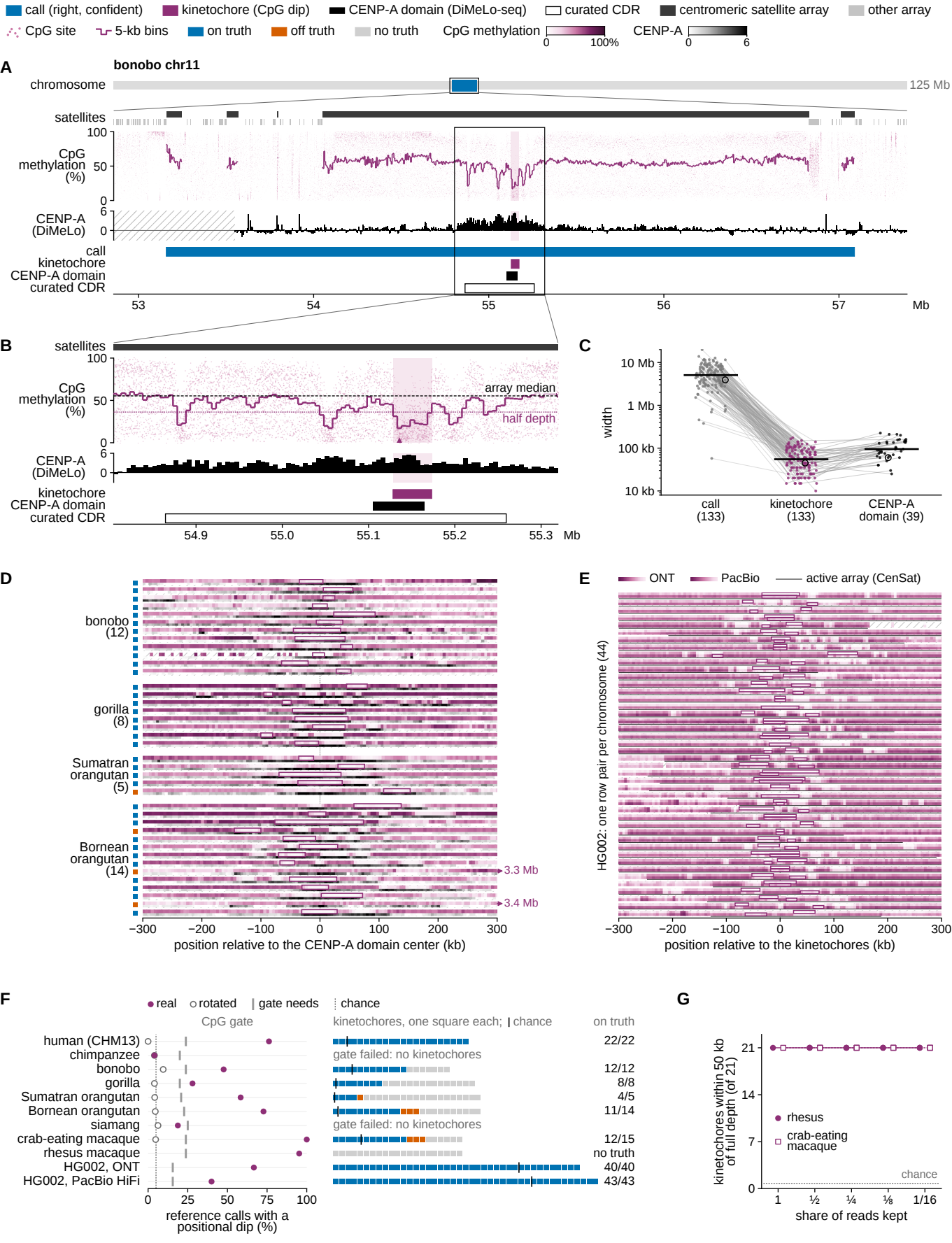

call (right, confident)
kinetochore (CpG dip)
CENP-A domain (DiMeLo-seq)
curated CDR
centromeric satellite array
other array
CpG site
5-kb bins
on truth
off truth
no truth
CpG methylation
0
100%
CENP-A
0
6
A
bonobo chr11
chromosome
125 Mb
satellites
CpG methylation (%)
CENP-A (DiMeLo)
call
kinetochore
CENP-A domain
curated CDR
53
54
55
56
57
Mb
B
array median
half depth
54.9
55.0
55.1
55.2
55.3
Mb
C
width
10 Mb
1 Mb
100 kb
10 kb
call (133)
kinetochore (133)
CENP-A domain (39)
D
bonobo (12)
gorilla (8)
Sumatran orangutan (5)
Bornean orangutan (14)
3.3 Mb
3.4 Mb
position relative to the CENP-A domain center (kb)
E
ONT
PacBio
active array (CenSat)
HG002: one row pair per chromosome (44)
position relative to the kinetochores (kb)
F
real
rotated
gate needs
chance
CpG gate
kinetochores, one square each; | chance
on truth
human (CHM13)
chimpanzee
bonobo
gorilla
Sumatran orangutan
Bornean orangutan
siamang
crab-eating macaque
rhesus macaque
HG002, ONT
HG002, PacBio HiFi
22/22
gate failed: no kinetochores
12/12
8/8
4/5
11/14
gate failed: no kinetochores
12/15
no truth
40/40
43/43
reference calls with a positional dip (%)
G
kinetochores within 50 kb of full depth (of 21)
rhesus
crab-eating macaque
chance
share of reads kept

Across primates the kinetochore sits in a CpG-hypomethylated dip a few tens of kilobases wide inside the megabase-long satellite array, so methylation from ordinary long reads locates CENP-A chromatin, and two sequencing technologies agree. Centromere identity leaves a mark that any long-read genome carries.

A A typical bonobo centromere. The satellite array spans megabases and its CpGs are about half methylated; in one place methylation drops, and that is where CENP-A, the histone that marks the kinetochore, is highest.

B The same place at the scale of single CpGs: the hypomethylated dip is tens of kilobases wide and sits inside the CENP-A domain, among shallower dips of the same array.

C The dip narrows the centromere from the megabase array to tens of kilobases, about the size of the CENP-A domain itself.

D Every ape kinetochore read out this way, aligned on its CENP-A domain: the light dip and the dark CENP-A signal line up on nearly every chromosome, in four species.

E The same diploid human genome read by two sequencing technologies gives dips in the same places.

F The dips sit at centromeres more often than methylation from elsewhere on the chromosome would put them, in most primates; where older data fail this test (chimpanzee, siamang), no kinetochore is placed. Wherever truth exists, most kinetochores land on it.

G A sixteenth of the reads, about one methylation call per CpG, still places every kinetochore in the same place.

Terms. Kinetochore: the CpG dip cenfuse reports inside its centromere call (v8 rerun; released CpG code): methylation pooled in 5-kb bins over arrays of the centromeric satellite family; the most prominent valley, extended to the bins below half its depth; shown where the genome's CpG gate passes and the dip beats the call's bins shuffled ($P \leq 0.05$). CENP-A domain: half-maximum run of DiMeLo-seq CENP-A signal (antibody-directed adenine methylation, independent of CpG calls). On truth: within 50 kb of the CENP-A domain (apes), overlapping the curated centromere dip region (CDR; human), inside the CENP-A ChIP domain (crab-eating macaque) or an active alpha-satellite array (HG002). Chance: a kinetochore of the same width placed at random inside its call. Inputs: public ONT CpG tracks of the assembled animals (apes R9, rhesus R10).

A Bonobo chr11 (CM055483.2), chosen by rule: of the 39 ape kinetochores with a CENP-A domain in their call, the one nearest the medians of kinetochore width (45 kb) and of the distance between kinetochore and CENP-A centers (23 kb) (next: gorilla CM068951.1, bonobo CM055476.2). Its call (3.93 Mb, confidence high) is the median of the 39. Top: the whole chromosome (125 Mb). Tracks: every satellite array (dark: the centromeric family, a 171-bp alpha satellite whose largest array is 2.76 Mb; light: others); every CpG of the input track (56,225 dots here) and the 5-kb bins cenfuse used (line); DiMeLo-seq CENP-A (hatched: no data); the call (blue: right, confident), the kinetochore (55.128-55.173 Mb), the CENP-A domain and the curated CDR. Pink band: the kinetochore. The array is methylated at 55% of its CpGs, the kinetochore at 24%.

B The boxed window of A. Dashed: array median of the 5-kb bins; dotted: half depth; triangle: the valley floor. The kinetochore (45 kb) overlaps the CENP-A domain (60 kb; centers 16 kb apart). The curated CDR (395 kb) spans several dips of similar depth; cenfuse reports the most prominent. Dip test $P = 0.005$ (200 shuffles), positional $P = 0.025$ (200 rotations).

C Widths, one point per chromosome; lines join a chromosome's call, kinetochore and CENP-A domain; bars: medians; circles: the example. All 133 kinetochores of the six genomes with a passing gate and truth (human 22, bonobo 19, gorilla 23, Sumatran orangutan 24, Bornean orangutan 24, crab-eating macaque 21): median 55 kb vs 5.1 Mb for their calls (92× narrower). The 39 with DiMeLo-seq truth: kinetochores 45 kb, calls 3.9 Mb, CENP-A domains 95 kb.

D One kinetochore per row pair: CpG methylation per 5-kb bin (upper strip; hatched: fewer than 10 CpGs) and DiMeLo-seq CENP-A (lower strip; one scale for all rows, clipped at 6), centered on the CENP-A domain, ± 300 kb; box: kinetochore; arrow: kinetochore outside the window; square: within 50 kb of CENP-A (blue) or not (vermilion). Rows by chromosome within species. 35 of 39 within 50 kb (29 overlap; chance 4.9, Poisson-binomial $P = 1.9 \times 10^{-34}$); per species: bonobo 12/12, gorilla 8/8, Sumatran orangutan 4/5, Bornean orangutan 11/14. CENP-A signal is higher inside the kinetochore than over its call on 39/39 (median 5.02 vs 0.81). Misses: two in the same array, Bornean orangutan CM054635.2 (50 kb, 1 bp beyond 50), Sumatran orangutan CM054703.2 (65 kb); two Bornean orangutan kinetochores lie in the large 171-bp array 3.3 and 3.4 Mb from CENP-A, which sits on small arrays of 181- and 189-bp families inside the same call. Chimpanzee has none: its gate failed.

E HG002 (diploid T2T, both haplotypes) with CpG methylation from ONT R10 (upper strip) and PacBio HiFi (lower strip) of the same cell line; boxes: each technology's kinetochore; line: CenSat active alpha-satellite arrays; rows centered on the mean of the two kinetochore midpoints, chr1 to chrY, maternal first (44 chromosomes with a kinetochore from either). The gate passed on 46 chromosomes with each. Kinetochores: ONT 40 (median 38 kb), PacBio 43 (25 kb), all wholly inside an active array (chance 30 and 32: calls lie on the arrays). Where both place one (39), they are within 50 kb on 31 (overlap 15); the other 8 lie 55-195 kb apart in the same array. Calls equal the FASTA-only run's. HG002 has no CENP-A data.

F Left, the CpG gate. A dip is positional when it is longer than dips made by the chromosome's own CpG values rotated onto the call ($P \leq 0.05$, 200 rotations). The gate passes when the other chromosomes' high-confidence calls hold more positional dips than 5% each would give (binomial $P \leq 0.01$; gray tick: dips needed). Filled: real tracks (v8 runs); open: every value moved half a chromosome along the same CpGs (v4 control runs, code cenfuse_cpg4, 2026-10-26; real tracks in that code pass and fail in the same genomes). Real: passes in 6/7 primates with truth, rhesus and HG002; fails in chimpanzee (1 of 25) and siamang (3 of 16), older R9 tracks. Rotated: 0-2 positional dips among the references; gate passes in 0/8 runs. Right: one square per kinetochore; black tick: chance. Human truth is the curated CDR, from the same reads (an implementation check); with the apes 62/63 overlap their CDR (Sumatran orangutan CM054694.2 is 2.3 Mb off). Crab-eating macaque: 12/15 inside the CENP-A ChIP domain of another animal; the other three lie 0.8-3.4 Mb from it inside the same call. Rhesus (held out of the design): 21 of 22 chromosomes with a kinetochore.

G The two tables with read counts (rhesus R10, about 15 calls per CpG strand; crab-eating macaque about 38) thinned binomially to 1/2 to 1/16 of the reads, 3 seeds each (v6 runs, code cenfuse_v6o: calls pooled per bin, the released rule; 2026-10-27); kinetochores within 50 kb of the full-depth run's. Rhesus at 1/16 (about one call per CpG strand): 21/21 in every seed; crab-eating macaque 21/21; chance at most 0.8 (dotted). No call changed.

Exceptions and caveats. The CpG line refines a call and never moves it: a wrong call gets a kinetochore inside the wrong array (Bornean orangutan CM054652.2, the confident ape miss, gets one in its inactive array, methylation 0.15 vs 0.32; absent from D, whose calls hold CENP-A). The CpG rules were designed on these primate tracks (not blind); rhesus was held out. Crab-eating macaque truth is from another animal. The README's width (55 kb, calls 5.1 Mb) is the median over all 133 kinetochores (C), not the 39 of D. The example is typical of all 39 by the rule above, not the cleanest: its CDR holds several dips and the kinetochore is one of them.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/update/figures/benchmark_cpg · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/update/figures/benchmark_cpg/benchmark_cpg.py · 2026-10-30